

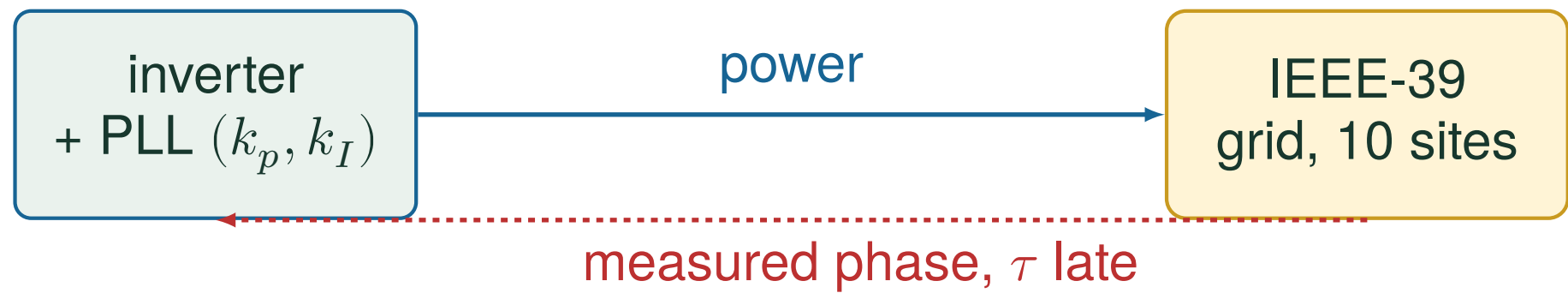
BEYOND NODAL DAMPING

PRESERVING ONE MODE DOES NOT PROTECT THE GRID

Local PLL retunes are exact; the network sets their price · IEEE-39, delayed nonlinear DAE
Jorge Luis Mayorga Taborda | Universidad de los Andes, Colombia

1 A PLL THAT LISTENS LATE

Inverters (GFL) replace synchronous generators. Each one tracks the grid phase with a phase-locked loop (PLL), and that measurement arrives τ late.



If a retune keeps one oscillation mode exactly where it was, is the grid safe?

IEEE-39 at 87.5 % inverter share, ten PLLs; the PLL delay grows from 40 to 44 ms. Full delayed DAE (204 states), exact exponentials.

HOW TO READ Left: the exact structure of a local retune (2–4). Centre: the main result, delay compensation (5–8). Right: what the network decides (9–12).

2 A RETUNE IS A RANK-ONE CHANGE

Each PLL enters the delayed grid only through its loop factor:

$$\Delta(s) = sI - A_0 - \sum_i b_i(k_i) c_i^\top e^{-s\tau_i}$$

Retuning a set S of m PLLs changes m columns, so the 204-state determinant reduces exactly to an $m \times m$ one:

$$\frac{\det \Delta_S(s)}{\det \Delta_0(s)} = \det(I_m + C_S^\top \Delta_0^{-1}(s) P_S)$$

EXACT Replacement plus retuning acts through at most 30 directions (10 at fixed share). Reconstruction error 3×10^{-16} over 21 designs and three delay fields.

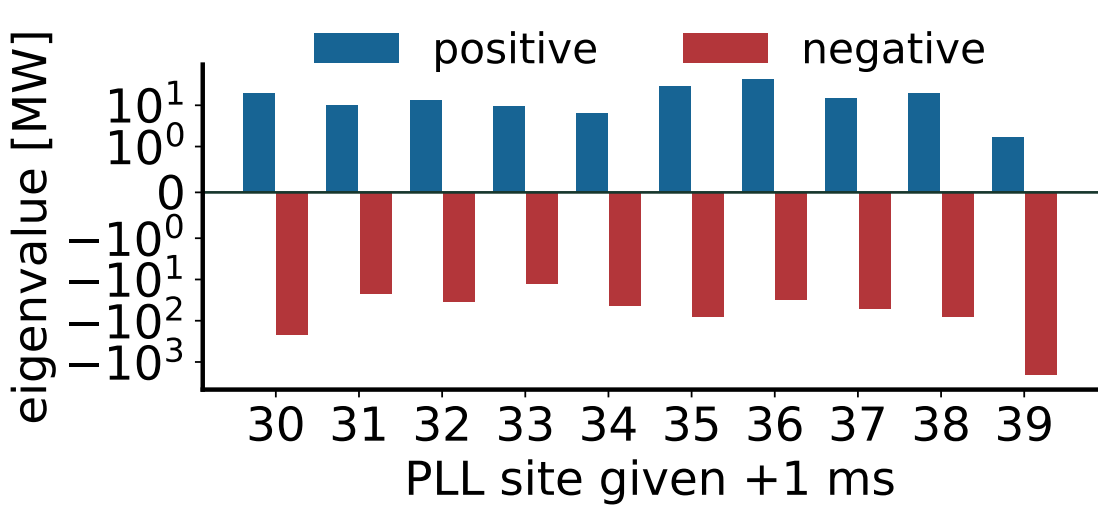
3 BEYOND NODAL: NEVER PURE DAMPING

Keep the generator speeds ν , eliminate the rest. The damping the grid supplies at frequency Ω is the Hermitian part of a full matrix, not a list of nodal coefficients:

$$Z = M(\Delta_{\nu\nu} - \Delta_{\nu h} \Delta_{hh}^{-1} \Delta_{h\nu}), \quad D_H = \frac{1}{2}(Z + Z^H)$$

One PLL change gives a rank-two increment with one positive and one negative eigenvalue:

$$\delta D_H = \frac{1}{2}(av^H + va^H), \quad \ell_+ \ell_- = -\frac{1}{4}(\|a\|^2 \|v\|^2 - |v^H a|^2)$$



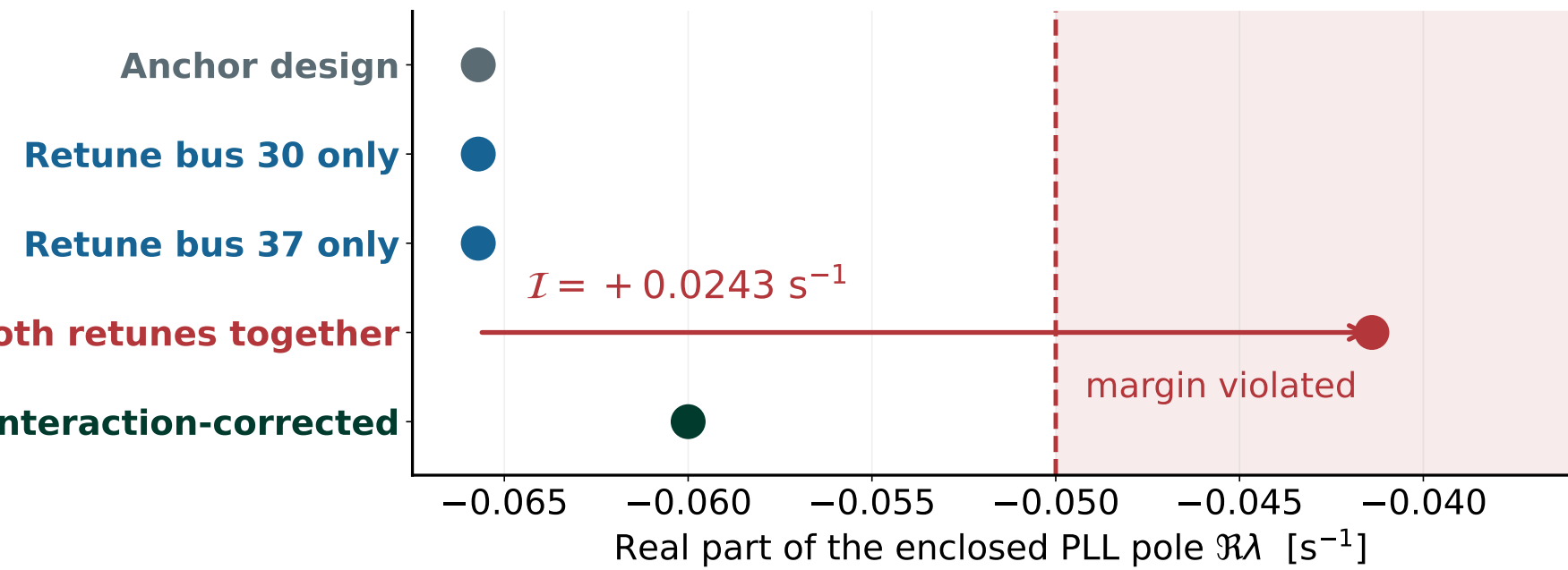
IEEE-39, +1 ms at each PLL, 5 Hz: 60 of 60 checks, error $< 5 \times 10^{-8}$.
One frozen design: nodal term +38.7, collective term -737.5.

COUNTING LAW Removing r anti-damped directions needs at least r retuned PLLs, for any gain size (nine-bus bank at 8 Hz: $r = 3$, interval-certified). Positive damping at one frequency is not stability.

4 SAFE ALONE CAN FAIL TOGETHER

Two retuned sites close a loop through the grid, $P = K_{ij}K_{ji}$ with $K_{ij} = (I + H_{ii})^{-1}H_{ij}$:

$$\frac{\det F \det F_0}{\det F_i \det F_j} = \det(I - P)$$



Buses 30 and 37 each hold the decay at -0.0657 s^{-1} ; together -0.0414 , past the -0.05 margin. Interaction $+0.0243 \pm 4 \times 10^{-7}$, interval-certified. A fixed-point budget restores -0.060 in 11 updates; 5 of 5 events.

Matching the full complex pole instead composes (0 of 230 failures).

5 TWO GAINS, ONE MODE: THE EXACT LAW

Loop factor $\kappa(s) = (k_p s + k_I) e^{-s\tau}$. The delay grows by h ; protect the mode $\lambda = \alpha + j\omega$.

Transport of the gains

$$k_p(h)\lambda + k_I(h) = (k_p\lambda + k_I) e^{\lambda h}$$

Error left at every other frequency

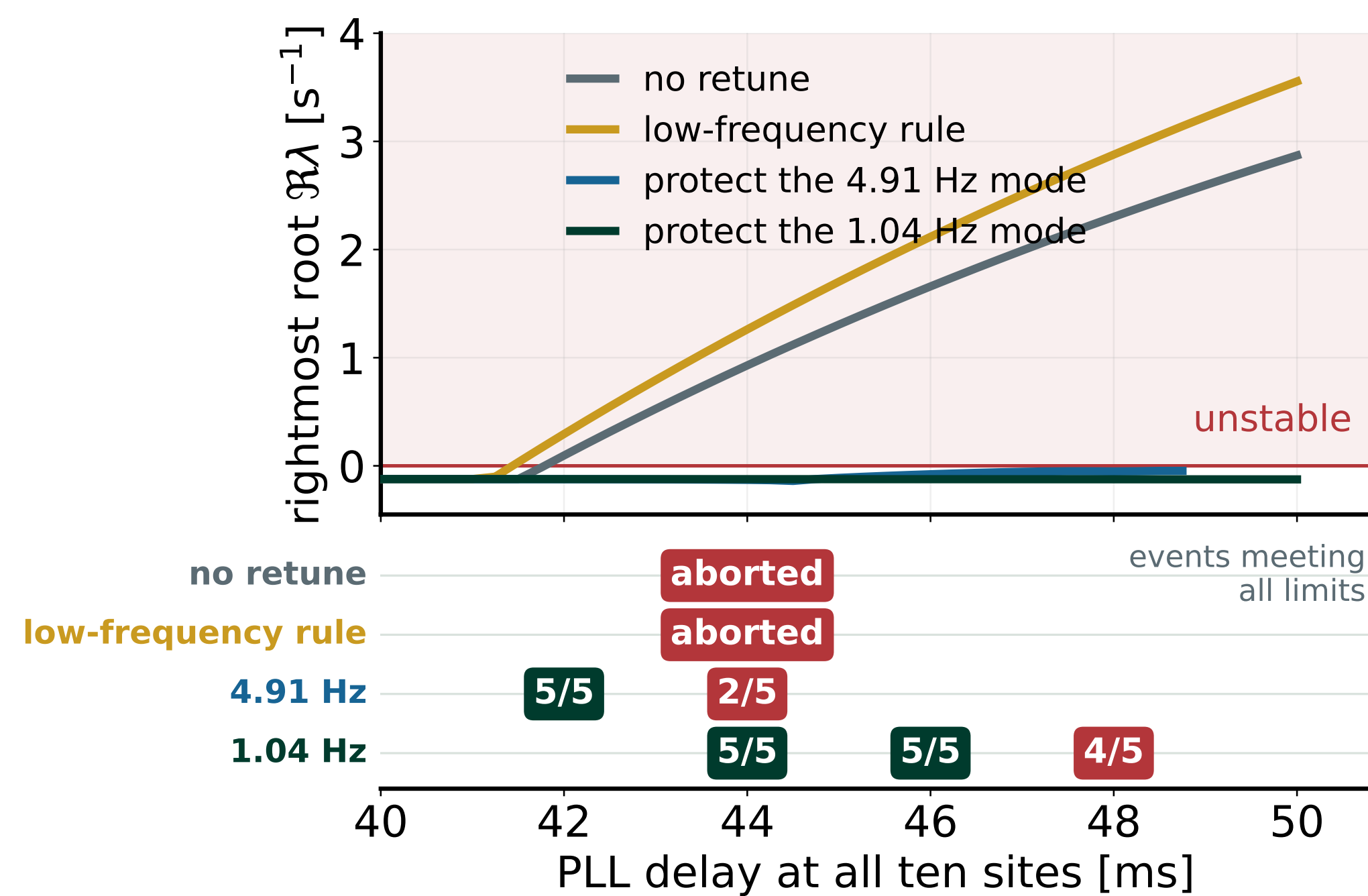
$$\Delta\kappa(s) = -(s - \lambda)(s - \bar{\lambda}) e^{-s\tau} \int_0^h k_p(t) e^{-st} dt$$

Spillover onto any other mode μ

$$\frac{d\mu}{dh} = -k_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial k_I}$$

IN WORDS The PLL fixes the quadratic factor; the network fixes the residue $\partial\mu/\partial k_I$. Gains need no network model; choosing λ does. Budget: $k_I > 0$ while $h < \varphi_{PI}(\omega)/\omega$.

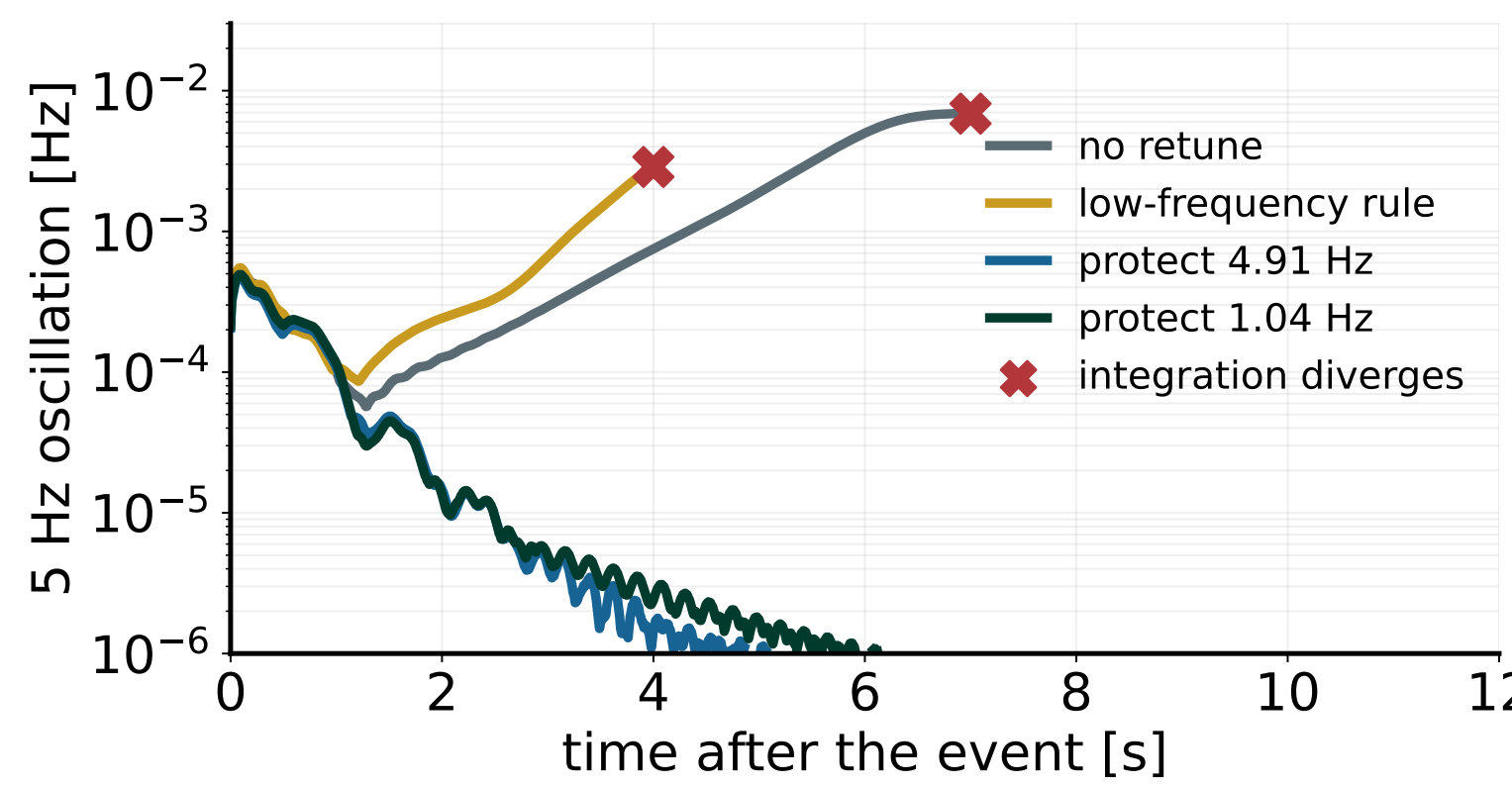
6 THE USUAL COMPENSATION MAKES IT WORSE



unstable roots at 44 ms: the low-frequency rule against protecting the 1.04 Hz mode (full-spectrum count).

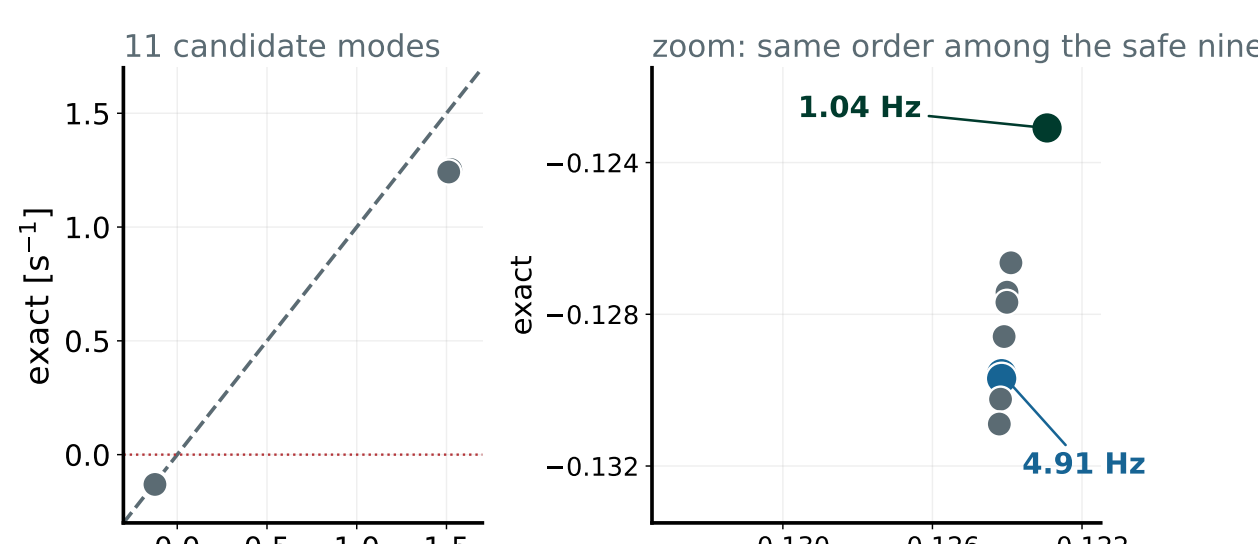
IN WORDS Which mode you protect decides everything. Protecting the 4.91 Hz mode keeps the spectrum but fails 3 of 5 events; protecting the 1.04 Hz mode passes 5 of 5.

7 IN TIME: WHAT THE GRID DOES



Bus 16 + 100 MW, 44 ms at all PLLs. 4.2–5.6 Hz band of the 39 bus frequencies. Without the right retune the 5 Hz oscillation grows until the integration diverges; protected, it dies out.

8 ONE ANALYSIS CHOOSES THE MODE



Predicted vs exact worst root after +4 ms, 11 candidate modes. Rank correlation 0.99: same order, not the same values.

The rule

- 1 One eigen-analysis of the base design.
- 2 Predict the worst root per candidate.
- 3 Among safe ones, keep the most k_I .
- 4 Verify roots and events.

9 SPECTRUM AND EVENTS: PASS BOTH

Roots clean, events fail

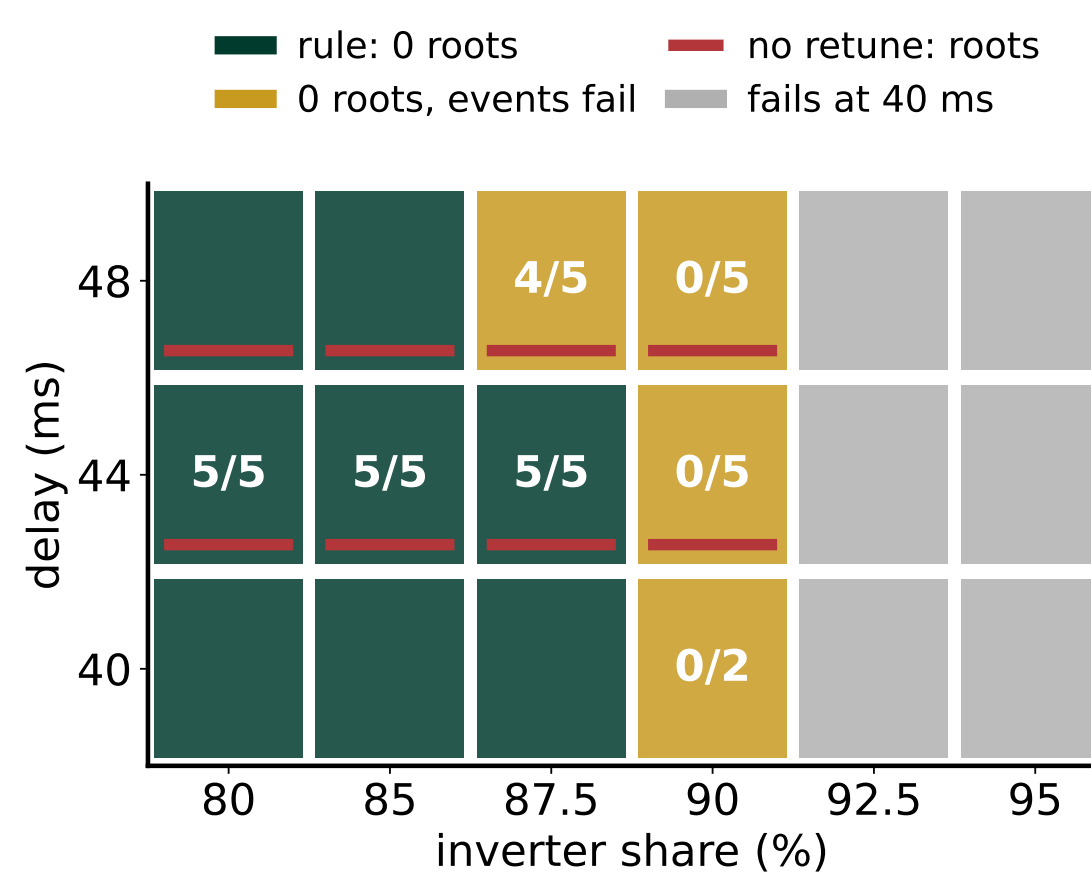
Protect 4.91 Hz at 44 ms: 0 roots beyond the margin, yet only **2 of 5** events meet the limits (0.505 Hz; SG reserve ≈ 0). The retune spends k_I : 246.7 $\rightarrow$ 137.8.

Events clean, roots unstable

No retune at 42 ms: 4 unstable roots ($+0.096 \text{ s}^{-1}$ at 4.8 Hz), yet the event meets every limit for 30 s. The 0.5 s windows miss slow 5 Hz growth.

Neither test is enough alone.

10 REACH AND TRANSFER



Each cell: roots beyond the margin with the rule; red bar: same cell, no retune; numbers: events passed.

IN WORDS The rule extends delay tolerance at every share where 40 ms is feasible. It does not raise the maximum share: at 90 % even the 40 ms design fails the events (0 of 2); 92.5 % fails at 40 ms.

Second design (85 %), rule applied unchanged: 0 roots, **5 of 5** events. **Nine-bus bank** (assumed data): $+0.886 \rightarrow -0.131 \text{ s}^{-1}$, re-run. **First-order formula**: same counts; the choice of mode matters, not the truncation.

11 WHICH SETS FAIL: PORTFOLIO CLOSURE

15/1

safe / unsafe subsets of $H_4 = \{30, 33, 35, 37\}$: every proper subset stable, the full set unstable ($+0.127 \text{ s}^{-1}$, 0.62 Hz).

Closure of a replacement set H through the network ports:

$$\det(I + Q_H) = \det(I + Q_{RR}) \det(I - R_{i|R})$$

$$R_{i|R} = Q_{iR}(I + Q_{RR})^{-1}Q_{Ri}$$

Adding bus 34 restores stability (-0.205 s^{-1}). Blind prediction from one network kernel: 16 of 16 on four buses; on nine buses 395 of 512 with 116 false-safe, so the closure explains but does not yet scale.

TX4 model with a fixed policy: no governor, no delay.

12 WHAT RETUNING CANNOT CHANGE

Low-frequency moments

With every k_I fixed, the first two moments of the frequency response stay put; any PLL only retune leaves the differences of bus frequency areas unchanged:

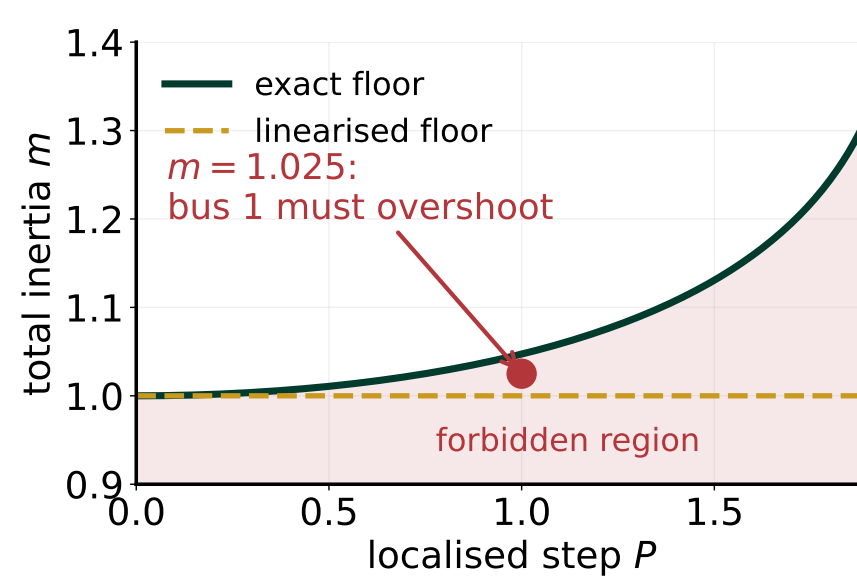
$$H_{\text{new}}(s) - H_{\text{base}}(s) = O(s^2), \quad \int_0^\infty (f_{\text{new}} - f_{\text{base}}) dt = 0$$

A network inertia floor

After a localised step, no local frequency avoids crossing its final value unless

$$m + \kappa > d_0^2 (e_k - c)^\top \bar{L}_k^\dagger (e_k - c)$$

Moving inertia or adding zero-net-energy damping cannot lower it. Two nodes: $\inf m = 2 \arcsin(P/2)/P$.



AUDIT Retracted: a 94.2 % inverter design fails RoCoF (0.513 Hz/s) and frequency (0.589 Hz). Invalid: zero-frequency Schur damping (condition 3×10^{18}). Certified: one root cluster over a full share/gain box (485 interval panels). Not claimed: maximum share, optimal mode, superiority over other compensators. Floor: reduced lossless model.

THREE RULES FOR THE ENGINEER 1 Do not compensate PLL delay at low frequency.

2 Choose the protected mode with the spillover law.

3 Check the spectrum and the events.

TAKEAWAY

PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

The gains that hide a delay from one mode are exact. The network decides what they do to all the others.

REFERENCES

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- [2] Huang et al., arXiv:1903.05489.
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Topics: [1] converter small-signal stability; [2] PLL stability and grid structure; [3] fixed-order design with delays; [4] sparse wide-area control; [5] nonlinear eigenvalues; [6] interval arithmetic.

CONTACT

Jorge Luis Mayorga Taborda
jl.mayorga@uniandes.edu.co
Universidad de los Andes
Bogotá, Colombia

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